

Cartaisy Mobile Latest SaaS-Readiness Status Report

Regenerated on June 30, 2026 after the original untracked PDF artifact was no longer present on disk. Read-only inspection report; no application code changes.

1. Executive Summary

Mobile SaaS-readiness has improved since the prior report. Production-grade public mobile config validation is implemented, CI now invokes typecheck/lint/test scripts, and cart, checkout, order list, and order detail flows now use shared unavailable-state handling for controlled store/catalog failures.

The biggest remaining gap is that dynamic native branding and runtime theme/logo branding are still audits/plans, not implemented behavior. The app still uses static app.json/native identity, static bundled logos, and hardcoded Tamagui theme tokens.

Final judgment: Mobile SaaS configuration safety is mostly complete but has known gaps. Confidence is medium-high for config validation and unavailable-state coverage; medium for branded build readiness because native build automation remains unsolved.

2. Recent Issues and PRs Reviewed

Area	PR/issue evidence	Status	Summary
Mobile CI/typecheck/test validation	PR #30, branch codex/issue-26-mobile-ci-validation	Partially completed	package.json and CI now run typecheck, lint, and test. Typecheck currently fails locally; test is still a placeholder.
Production-grade mobile config validation	PR #31, branch codex/mobile-config-validation	Completed for current phase	api/config/mobileConfig.ts validates required API base URL and Store ID, forbidden EXPO_PUBLIC_SHOPIFY_* variables, and secret-looking public keys.
Dynamic app config/native identity audit	PR #32, issue #27	Completed as audit only	docs/DYNAMIC_APP_CONFIG_MIGRATION_AUDIT.md documents static app.json risks and recommended app.config.ts migration.
Cart/checkout/orders unavailable states	PR #33, issue #29	Mostly completed	Cart, checkout, shipping, confirmation, orders list, and order detail now surface controlled store/catalog unavailable failures.
Runtime branding contract audit	PR #35, issue #34	Completed as audit only	docs/MOBILE_RUNTIME_BRANDING_CONTRACT.md proposes optional branding fields on /store/config. Runtime branding is not implemented yet.

3. Mobile Config Validation Status

Question	Current answer
Missing API base URL detected?	Yes. EXPO_PUBLIC_API_BASE_URL is required.
Invalid API base URL detected?	Yes. It must be an absolute HTTP/HTTPS URL with a hostname.
Missing Store ID detected?	Yes. EXPO_PUBLIC_STORE_ID is required.
Invalid Store ID detected?	Yes. It must match a 24-character Mongo ObjectId pattern.
Forbidden EXPO_PUBLIC_SHOPIFY_* variables?	Yes. Known keys are statically referenced for Expo bundle detection and all matching keys are flagged.
Secret-looking EXPO_PUBLIC_* variables?	Yes. SECRET, PASSWORD, PRIVATE_KEY, ACCESS_TOKEN, REFRESH_TOKEN, ADMIN_TOKEN, and CLIENT_SECRET patterns are flagged.
Development behavior	Logs an error and keeps local boot possible. Invalid required config values are exposed as empty strings to prevent accidental requests.
Production behavior	Throws before API requests, giving fail-fast behavior for shipped builds.
Tests	No dedicated validator tests were found.

Local environment note: lint output showed a local .env exporting EXPO_PUBLIC_SHOIFY_* variable names. Values were not inspected or copied, but those names should be removed from mobile build environments.

4. Mobile CI / Typecheck / Test Status

Check	Status
package.json typecheck script	Yes: tsc --noEmit.
CI runs typecheck	Yes, but local npm run typecheck currently fails with existing TypeScript errors.
CI runs lint	Yes. Local npm run lint completed with 0 errors and 112 warnings.
CI runs tests	Yes, but npm test only echoes "No mobile test runner configured yet". No real tests execute.
Dependency installation	CI uses npm ci with package-lock.json on Node 20.
Recommended local commands	npm run typecheck, npm run lint, npm test. Typecheck must be fixed for reliable CI.

5. Dynamic Branded App Config Status

Dynamic app config is not implemented. The repository still uses static app.json and no app.config.ts/app.config.js is checked in.

Surface	Status
App name / slug / scheme	Static cartaisy values in app.json/native files.
iOS bundle ID	Mismatch risk: app.json and GoogleService-Info.plist use com.rendernext.cartaisy, while checked-in Xcode project uses com.cartaisy.app.
Android package	Consistent in inspected files as com.rendernext.cartaisy.
Firebase	Static per-repo files; iOS Firebase config matches app.json, not the checked-in Xcode bundle ID.
Stripe / Apple Pay	Static merchant.com.cartaisy; checked-in iOS entitlements do not include Apple Pay entitlement and aps-environment is development.
Remaining risk	Per-merchant native builds remain manual/high-risk until app.config.ts and build automation are implemented.

6. Runtime Branding Status

Runtime branding is documented but not implemented. GET /store/config currently provides currency, timezone, and name. Store config is persisted as currency, timezone, and storeName.

Item	Status
Theme colors	Still hardcoded in tamagui/token.ts with PRIMARY_COLOR #A82A50.
Logo	Still bundled through assets/lcons.ts as cartaisyColorlogo and cartaisyWhitelogo.
/store/config branding	Not consumed in active code. Audit proposes optional branding.shortName, primaryColor, secondaryColor, logoUrl, logoDarkUrl, and updatedAt/version.
Fallback behavior	Recommended in docs but not implemented. ApplImage can load remote images, but remote image error state is not used for fallback yet.
Backend requirement	Add optional public branding fields, validate values, and return public HTTPS logo URLs only.

7. Store/Catalog Unavailable Coverage

Flow	Status	User sees / retry	Risk
Home catalog	Implemented	Catalog unavailable state with retry/refetch.	Low
Product listing / collection	Implemented	Catalog unavailable state with retry.	Low
Product detail	Implemented	Catalog/Product unavailable state with retry.	Low

Flow	Status	User sees / retry	Risk
Search	Implemented	Catalog unavailable state with retry.	Low
Cart	Mostly implemented	Cart unavailable full-screen state; retry syncs cart.	Medium
Checkout	Mostly implemented	Checkout unavailable full-screen state; retry clears unavailable state and user may need to retry action.	Medium
Orders list	Implemented	Orders unavailable state with refetch.	Low
Order detail	Implemented	Order unavailable state with refetch.	Low
Wishlist	Unclear / partial	No focused wishlist unavailable audit found.	Medium

Controlled errors include store not found, store inactive, store unavailable, store configuration errors, missing Shopify credentials/config, Shopify unavailable/configuration errors, catalog unavailable, and matching 403/423/503 payloads with store/Shopify/catalog context.

8. Unsafe Shopify Credential Status

Question	Answer
Is EXPO_PUBLIC_SHOPIFY_ACCESS_TOKEN removed from active API use?	Yes. It is only referenced in mobileConfig.ts for forbidden-key detection and appears in docs/history.
Any EXPO_PUBLIC_SHOPIFY_* in active config/code?	Only detection code. A local .env has forbidden names and should be cleaned.
Does mobile directly call Shopify APIs?	No direct Shopify API credential calls were found; active flows use Cartaisy backend endpoints.
Any Shopify Admin/private tokens exposed?	No committed active token values were found. Local secret values were not inspected.

9. Current Readiness Matrix

Area	Status	Remaining gap	Risk
AI workflow rules / PR template	Complete	Keep enforcing issue-linked focused PRs.	Low
CI	Partial	Typecheck fails and test is no-op.	High
Shopify credential cleanup	Mostly complete	Remove local/public Shopify env names from mobile build environments.	High
Required config validation	Complete for current phase	Add automated tests.	Medium
Dynamic native app identity	Partial	Implement app.config.ts and per-merchant build inputs.	High
Runtime theme/logo	Not started in code	Implement /store/config branding and fallbacks.	Medium
Catalog/cart/checkout/orders unavailable UI	Mostly complete	Add tests and verify edge cases manually.	Medium
Per-merchant build automation	Not started	Automate assets, Firebase, package IDs, signing inputs, and validation.	High
Preview app flow	Unclear / not started	Define tenant-safe preview model.	High

10. Remaining Mobile Risks

Open risks include Shopify-hosted checkout strategy, native build automation per merchant, app icon/splash generation, bundle ID/package automation, Firebase config per merchant, Stripe/Apple Pay merchant config per merchant, preview app flow, OTA/update strategy, runtime theme/logo implementation, missing validation tests, failing typecheck, no real test runner, and app review/submission readiness.

11. Final Judgment

Mobile SaaS configuration safety is mostly complete but has known gaps. The current phase has strong config validation and broader unavailable-state handling, but it is not fully complete until native app identity, runtime branding, build automation, typecheck, and tests are addressed.

12. Next Recommended Mobile Issues

Priority	Title	Why it matters	Owner / risk
1	Fix mobile typecheck gate and generated API type mismatches	CI cannot be trusted until typecheck passes.	AI can help; high risk.
2	Implement dynamic app.config.ts for branded mobile builds	Merchant-specific native identity must be automated and validated.	Human leads decisions; AI can implement; high risk.
3	Add real mobile config validation tests	Protects production fail-fast and forbidden public secret detection.	AI can implement; medium risk.
4	Implement runtime mobile branding from /store/config	Enables merchant logo/colors without native rebuild for safe in-app surfaces.	AI can implement after backend contract; medium risk.
5	Design per-merchant build and preview workflow	Defines production build inputs, preview trust model, signing, Firebase/payment ownership, and OTA strategy.	Human should lead; high risk.

Commands Verified

Command	Result
npm run typecheck	Failed with existing TypeScript errors.
npm run lint	Passed with 0 errors and 112 warnings.
npm test	Passed only because no test runner is configured.
PDF file check	Regenerated PDF exists and begins with a valid %PDF header.